

PROFESSIONAL CURRICULUM

CEH v13 / CEH AI-Aligned Professional Cybersecurity & Ethical Hacking Curriculum

Ethical Hacking | Offensive Security | Defensive Security
SOC & SIEM | Threat Hunting | DFIR | Cloud | DevSecOps | AI Security

20

CEH-aligned core modules

12

professional cybersecurity modules

587

structured lessons /
video topics

Independent curriculum outline. Use all offensive-security activities only with explicit authorization.

Curriculum Overview

This curriculum combines a CEH v13 / CEH AI-aligned ethical-hacking core with a broader professional cybersecurity track. It is designed to progress from safe lab setup and foundational security concepts through offensive assessment, defensive engineering, security operations, incident response, cloud and container security, secure development, and AI security.

Structure	Coverage	Delivery emphasis
20 CEH-aligned core modules + 12 professional cybersecurity modules	Ethical hacking, network/web/mobile/IoT/cloud security, GRC, IAM, SOC/SIEM, threat hunting, DFIR, DevSecOps and AI security	Theory, authorized labs, detection and defense, reporting, capstones and career development

Safety and authorization: All attack simulation, exploitation, password auditing, wireless testing, malware analysis, DoS simulation, social-engineering simulation, and related offensive activities in this curriculum are intended only for systems you own or have explicit written authorization to test, preferably inside isolated lab environments.

Positioning: This is an independent professional curriculum outline. It is not presented as official EC-Council courseware or an EC-Council-authorized training program unless the training provider separately holds the required authorization.

Table of Contents

The curriculum is divided into the CEH-aligned core and an extended professional cybersecurity track. Page numbers are generated from the final layout.

Curriculum Overview	2
PART I - CEH v13 / CEH AI-ALIGNED CORE	5
0. COURSE INTRODUCTION	5
1. CYBERSECURITY LAB SETUP	5
MODULE 01 - INTRODUCTION TO ETHICAL HACKING & CYBERSECURITY	5
MODULE 02 - FOOTPRINTING, RECONNAISSANCE & OSINT	6
MODULE 03 - SCANNING NETWORKS	6
MODULE 04 - ENUMERATION	6
MODULE 05 - VULNERABILITY ANALYSIS & MANAGEMENT	7
MODULE 06 - SYSTEM HACKING & SYSTEM SECURITY	7
MODULE 07 - MALWARE THREATS & MALWARE DEFENSE	8
MODULE 08 - SNIFFING & NETWORK TRAFFIC SECURITY	8
MODULE 09 - SOCIAL ENGINEERING & SECURITY AWARENESS	8
MODULE 10 - DENIAL-OF-SERVICE & AVAILABILITY SECURITY	9
MODULE 11 - SESSION HIJACKING & SESSION SECURITY	9
MODULE 12 - IDS, IPS, FIREWALLS, HONEYPOTS & NETWORK DEFENSE	9
MODULE 13 - WEB SERVER SECURITY	10
MODULE 14 - WEB APPLICATION & API SECURITY	10
MODULE 15 - SQL INJECTION & DATABASE SECURITY	11
MODULE 16 - WIRELESS NETWORK SECURITY	11
MODULE 17 - MOBILE APPLICATION & MOBILE DEVICE SECURITY	11
MODULE 18 - IoT, IIoT, OT, ICS & SCADA SECURITY	12
MODULE 19 - CLOUD, CONTAINER & KUBERNETES SECURITY	12
MODULE 20 - CRYPTOGRAPHY & DATA SECURITY	13
PART II - PROFESSIONAL CYBERSECURITY TRACK	14
MODULE 21 - CYBERSECURITY GOVERNANCE, RISK & COMPLIANCE	14
MODULE 22 - NETWORK SECURITY & SECURE INFRASTRUCTURE	14
MODULE 23 - IDENTITY & ACCESS MANAGEMENT	15
MODULE 24 - WINDOWS, LINUX & ENDPOINT SECURITY	15
MODULE 25 - SOC, SIEM & SECURITY MONITORING	16
MODULE 26 - CYBER THREAT INTELLIGENCE & THREAT HUNTING	16
MODULE 27 - INCIDENT RESPONSE & DIGITAL FORENSICS	17
MODULE 28 - DATA SECURITY, BACKUP, DISASTER RECOVERY & BUSINESS CONTINUITY	17
MODULE 29 - SECURE DEVELOPMENT, DEVSECOPS & SUPPLY-CHAIN SECURITY	18

MODULE 30 - AI & MACHINE LEARNING SECURITY 18

MODULE 31 - PENETRATION TESTING, PURPLE TEAM & SECURITY ASSESSMENT 19

MODULE 32 - CAPSTONE, REPORTING, EXAM PREPARATION & CAREER DEVELOPMENT 19

Alignment References & Authoritative Sources 21

Document Notes 21

PART I - CEH v13 / CEH AI-ALIGNED CORE

Foundational cybersecurity, ethical hacking methodology, reconnaissance, scanning, enumeration, vulnerability analysis, system security, malware, network traffic analysis, social engineering, availability, session security, network defense, web and API security, database security, wireless, mobile, IoT/OT, cloud, and cryptography.

0. COURSE INTRODUCTION

6 lessons

- 0.1 Introduction to CEH v13 / CEH AI & Professional Cybersecurity Course
- 0.2 Ethical Hacking Ethics, Rules, Authorization, Scope & Lab Safety
- 0.3 Cybersecurity vs Ethical Hacking vs Penetration Testing vs Red Teaming
- 0.4 Blue Team, Red Team, Purple Team & Security Operations
- 0.5 What's New in CEH v13 / CEH AI - AI, Modern Attacks & Emerging Technologies
- 0.6 Cybersecurity Career Paths, Roles & Learning Roadmap

1. CYBERSECURITY LAB SETUP

10 lessons

- 1.1 Virtualization Concepts & Installing VMware Workstation
- 1.2 Enabling Virtualization Technology - Intel VT-x & AMD-V
- 1.3 Installing Kali Linux on VMware
- 1.4 Kali Linux Repository, User, Updates & System Setup
- 1.5 Installing Windows Client VM for Cybersecurity Labs
- 1.6 Windows Server & Active Directory Lab Introduction
- 1.7 Installing Vulnerable Practice Systems - Metasploitable, DVWA & Others
- 1.8 Creating an Isolated Ethical Hacking & Cybersecurity Lab
- 1.9 VM Networking - NAT, Bridged, Host-Only & Internal Networks
- 1.10 VM Snapshots, Restore Points & Safe Lab Management

MODULE 01 - INTRODUCTION TO ETHICAL HACKING & CYBERSECURITY

15 lessons

- M01.1 Introduction to Cybersecurity, Ethical Hacking & Information Security
- M01.2 CIA Triad - Confidentiality, Integrity & Availability
- M01.3 Authentication, Authorization, Accounting & Non-Repudiation
- M01.4 Threats, Vulnerabilities, Risks, Exploits & Attack Surfaces
- M01.5 Hacker Classes, Attack Types & Ethical Hacking Methodology
- M01.6 Five Phases of Ethical Hacking
- M01.7 Cyber Kill Chain
- M01.8 MITRE ATT&CK Framework
- M01.9 Threat Intelligence Fundamentals
- M01.10 Security Controls - Administrative, Technical & Physical
- M01.11 Defense-in-Depth & Layered Security
- M01.12 Risk Management Fundamentals
- M01.13 Cybersecurity Laws, Regulations, Policies & Compliance
- M01.14 AI-Driven Ethical Hacking & Generative AI for Cybersecurity
- M01.15 Responsible & Secure Use of AI in Cybersecurity

MODULE 02 - FOOTPRINTING, RECONNAISSANCE & OSINT

14 lessons

- M02.1 Information Gathering, Footprinting & Reconnaissance
- M02.2 Passive vs Active Reconnaissance
- M02.3 Search Engine Reconnaissance & Google Hacking
- M02.4 WHOIS & Domain Footprinting
- M02.5 DNS Footprinting & DNS Records
- M02.6 Email & Website Footprinting
- M02.7 Social Media & People Search OSINT
- M02.8 SpiderFoot Automated OSINT
- M02.9 theHarvester OSINT & Email/Domain Reconnaissance
- M02.10 Shodan & Internet-Connected Device Reconnaissance
- M02.11 Metadata & Document Intelligence
- M02.12 OSINT Verification & Information Validation
- M02.13 AI-Powered OSINT & Reconnaissance
- M02.14 Reconnaissance Detection & Countermeasures

MODULE 03 - SCANNING NETWORKS

12 lessons

- M03.1 Network Scanning Fundamentals
- M03.2 TCP/IP, Ports, Protocols & Services for Security Testing
- M03.3 Nmap Host Discovery
- M03.4 Nmap TCP & UDP Port Scanning
- M03.5 Nmap Service & Version Detection
- M03.6 Operating System Detection & Banner Grabbing
- M03.7 Nmap Scripting Engine - NSE
- M03.8 Masscan & High-Speed Network Discovery
- M03.9 Network Mapping & Service Identification
- M03.10 Scanning Beyond Firewalls and IDS - Concepts & Detection
- M03.11 AI-Assisted Network Scan Analysis
- M03.12 Scan Detection, Prevention & Countermeasures

MODULE 04 - ENUMERATION

13 lessons

- M04.1 Introduction to Enumeration
- M04.2 SMB & NetBIOS Enumeration
- M04.3 SNMP Enumeration
- M04.4 LDAP & Active Directory Enumeration
- M04.5 DNS Enumeration
- M04.6 NFS Enumeration
- M04.7 SMTP Enumeration
- M04.8 RPC & FTP Enumeration
- M04.9 Windows Enumeration
- M04.10 Linux & Unix Enumeration
- M04.11 Active Directory Objects, Users, Groups & Trusts
- M04.12 AI-Assisted Enumeration

M04.13 Enumeration Detection & Countermeasures**MODULE 05 - VULNERABILITY ANALYSIS & MANAGEMENT**

15 lessons

- M05.1** Vulnerabilities, Weaknesses, Exploits & Misconfigurations
- M05.2** CVE - Common Vulnerabilities and Exposures
- M05.3** CVSS Vulnerability Scoring
- M05.4** NVD & Vulnerability Databases
- M05.5** Vulnerability Management Lifecycle
- M05.6** Vulnerability Scanning vs Vulnerability Assessment vs Penetration Testing
- M05.7** Web Server Vulnerability Scanning with Nikto
- M05.8** Vulnerability Assessment with Greenbone OpenVAS
- M05.9** Vulnerability Assessment with Nessus
- M05.10** Rapid7 InsightVM Vulnerability Management
- M05.11** False Positives, False Negatives & Vulnerability Validation
- M05.12** Risk-Based Vulnerability Prioritization
- M05.13** Patch Management & Remediation
- M05.14** AI-Powered Vulnerability Analysis
- M05.15** AI-Assisted Vulnerability Prioritization & Reporting

MODULE 06 - SYSTEM HACKING & SYSTEM SECURITY

22 lessons

- M06.1** System Hacking Methodology
- M06.2** Password Security & Authentication Fundamentals
- M06.3** Password Hashes & Password Storage
- M06.4** Password Auditing with John the Ripper
- M06.5** Password Auditing with Hashcat
- M06.6** Authentication Security Testing with Hydra in an Authorized Lab
- M06.7** Vulnerability Exploitation with Metasploit Framework
- M06.8** Windows Privilege Escalation Concepts
- M06.9** Linux Privilege Escalation Concepts
- M06.10** Active Directory Security Fundamentals
- M06.11** Active Directory Enumeration & Security with BloodHound
- M06.12** Credential Security & Credential Attack Concepts
- M06.13** Lateral Movement Concepts & Defensive Detection
- M06.14** Buffer Overflow Fundamentals
- M06.15** Buffer Overflow Detection & Defense
- M06.16** Persistence & Post-Exploitation Concepts
- M06.17** Persistence Detection & Security Monitoring
- M06.18** Steganography, Steganalysis & Data Hiding
- M06.19** Covering Tracks, Log Security & Detection
- M06.20** Windows Security Hardening Basics
- M06.21** Linux Security Hardening Basics
- M06.22** AI-Assisted System Security Assessment

MODULE 07 - MALWARE THREATS & MALWARE DEFENSE

17 lessons

- M07.1 Malware Fundamentals
- M07.2 Trojans, Viruses & Worms
- M07.3 Ransomware & Modern Extortion Threats
- M07.4 Rootkits, Spyware, Keyloggers & Botnets
- M07.5 APTs & Advanced Malware
- M07.6 Fileless Malware & Living-off-the-Land Techniques
- M07.7 AI-Generated & AI-Assisted Malware Threats
- M07.8 Safe Static Malware Analysis
- M07.9 Dynamic Malware Analysis in an Isolated Lab
- M07.10 Malware Analysis with Ghidra
- M07.11 Online Malware Analysis & Sandbox Services
- M07.12 Indicators of Compromise - IOC
- M07.13 YARA Fundamentals for Malware Detection
- M07.14 Antivirus, EDR & XDR Fundamentals
- M07.15 Malware Detection, Prevention & Countermeasures
- M07.16 Ransomware Prevention, Detection & Recovery
- M07.17 AI-Assisted Malware Analysis & Detection

MODULE 08 - SNIFFING & NETWORK TRAFFIC SECURITY

12 lessons

- M08.1 Packet Sniffing & Network Traffic Analysis
- M08.2 Ethernet, TCP/IP & Packet Structure Review
- M08.3 Wireshark Network Traffic Analysis
- M08.4 Wireshark Display & Capture Filters
- M08.5 ARP Spoofing & MITM Concepts
- M08.6 Ettercap & Bettercap in an Authorized Lab
- M08.7 DHCP Security
- M08.8 MAC Spoofing & MAC Security
- M08.9 VLAN & STP Security Concepts
- M08.10 DNS Security & DNS Attack Concepts
- M08.11 Detecting ARP Poisoning & MITM Attacks
- M08.12 Network Sniffing Detection & Countermeasures

MODULE 09 - SOCIAL ENGINEERING & SECURITY AWARENESS

15 lessons

- M09.1 Social Engineering Fundamentals
- M09.2 Psychology & Human Factors in Cybersecurity
- M09.3 Phishing & Spear Phishing
- M09.4 Impersonation & Pretexting
- M09.5 Smishing - SMS Phishing
- M09.6 Vishing - Voice Phishing
- M09.7 Identity Theft & Credential Theft
- M09.8 Mobile Social Engineering
- M09.9 Controlled Phishing Simulation & Awareness Training

- M09.10 AI-Generated Phishing
- M09.11 Deepfake Voice & Video Threats
- M09.12 Business Email Compromise - BEC
- M09.13 Phishing Detection & Email Security
- M09.14 Social Engineering Detection & Countermeasures
- M09.15 Building a Cybersecurity Awareness Program

MODULE 10 - DENIAL-OF-SERVICE & AVAILABILITY SECURITY

10 lessons

- M10.1 DoS & DDoS Fundamentals
- M10.2 Botnets & Distributed Attacks
- M10.3 Network-Layer DoS Concepts
- M10.4 Application-Layer DoS Concepts
- M10.5 Safe DoS Traffic Simulation in an Isolated Lab
- M10.6 DoS/DDoS Detection with Wireshark
- M10.7 Detecting DoS with IDS & Monitoring Systems
- M10.8 DDoS Mitigation & Rate Limiting
- M10.9 WAF, CDN & Cloud DDoS Protection
- M10.10 Availability, Resilience & Redundancy

MODULE 11 - SESSION HIJACKING & SESSION SECURITY

11 lessons

- M11.1 Session Management Fundamentals
- M11.2 Session Hijacking Concepts
- M11.3 Cookies, Tokens & Session IDs
- M11.4 Secure Cookie Attributes
- M11.5 Application-Level Session Attacks
- M11.6 Network-Level Session Attack Concepts
- M11.7 Session Fixation
- M11.8 Token Security
- M11.9 JWT Security Fundamentals
- M11.10 Session Security Testing in an Authorized Web Lab
- M11.11 Detecting & Preventing Session Hijacking

MODULE 12 - IDS, IPS, FIREWALLS, HONEYPOTS & NETWORK DEFENSE

16 lessons

- M12.1 Network Security Architecture
- M12.2 Firewalls & Firewall Types
- M12.3 IDS vs IPS
- M12.4 Network Access Control - NAC
- M12.5 Endpoint Security Architecture
- M12.6 IDS/Firewall Evasion Concepts
- M12.7 Evasion Detection & Defensive Countermeasures
- M12.8 Snort IDS Fundamentals
- M12.9 Suricata IDS/IPS Fundamentals
- M12.10 IDS Rules & Alert Fundamentals

- M12.11 Honeypots & Honeynets
- M12.12 Cowrie Honeypot Lab
- M12.13 Analyzing Honeypot Security Events
- M12.14 Network Segmentation & Security Zones
- M12.15 DMZ Architecture
- M12.16 Zero Trust Architecture Fundamentals

MODULE 13 - WEB SERVER SECURITY

12 lessons

- M13.1 Web Server Architecture
- M13.2 HTTP & HTTPS Fundamentals
- M13.3 Web Server Threats & Vulnerabilities
- M13.4 Web Server Reconnaissance
- M13.5 Web Server Enumeration
- M13.6 Apache Security Assessment
- M13.7 Nginx Security Assessment
- M13.8 Web Server Vulnerability Scanning
- M13.9 TLS/SSL Configuration Security
- M13.10 Web Server Logging & Monitoring
- M13.11 Web Server Hardening
- M13.12 AI-Assisted Web Server Security Assessment

MODULE 14 - WEB APPLICATION & API SECURITY

23 lessons

- M14.1 Web Application Architecture & Security Fundamentals
- M14.2 HTTP Requests, Responses, Headers & Methods
- M14.3 OWASP Top 10 Web Application Risks
- M14.4 Burp Suite Community Edition
- M14.5 OWASP ZAP Web Application Testing
- M14.6 DVWA Vulnerable Web Application Lab
- M14.7 WebGoat Security Lab
- M14.8 OWASP Juice Shop Security Lab
- M14.9 Web Application Reconnaissance
- M14.10 Web Application Vulnerability Scanning
- M14.11 Authentication Security Testing
- M14.12 Authorization & Access-Control Testing
- M14.13 IDOR & Broken Access Control Concepts
- M14.14 File Upload Security
- M14.15 Input Validation Security
- M14.16 Web Application Fuzzing Fundamentals
- M14.17 REST API Security Fundamentals
- M14.18 Webhooks & Modern API Architecture
- M14.19 OWASP API Security Risks
- M14.20 API Authentication & Authorization Security
- M14.21 Secure Headers & Browser Security Controls
- M14.22 Web Application Firewall - WAF Fundamentals

M14.23 AI-Assisted Web Application Security Testing

MODULE 15 - SQL INJECTION & DATABASE SECURITY

13 lessons

- M15.1 Database & SQL Fundamentals for Cybersecurity
- M15.2 SQL Injection Fundamentals
- M15.3 Error-Based SQL Injection
- M15.4 UNION-Based SQL Injection
- M15.5 Blind SQL Injection
- M15.6 SQL Injection Testing in an Authorized Lab
- M15.7 DVWA SQL Injection Practical
- M15.8 Automated SQL Injection Assessment with sqlmap
- M15.9 Database Authentication & Authorization Security
- M15.10 Parameterized Queries & Secure Database Development
- M15.11 SQL Injection Detection & Prevention
- M15.12 Database Logging & Monitoring
- M15.13 AI-Assisted SQL Injection Analysis

MODULE 16 - WIRELESS NETWORK SECURITY

15 lessons

- M16.1 Wireless Networking Fundamentals
- M16.2 Wi-Fi Standards & Wireless Architecture
- M16.3 WPA2 Security
- M16.4 WPA3 Security
- M16.5 Wireless Discovery with Kismet
- M16.6 Wireless Traffic Analysis
- M16.7 Aircrack-ng Wireless Security Auditing
- M16.8 WPA/WPA2 Handshake Analysis in an Authorized Lab
- M16.9 Wireless Password Security Assessment
- M16.10 Rogue Access Points
- M16.11 Evil Twin Attack Concepts
- M16.12 Wireless Deauthentication Security Concepts
- M16.13 Wireless Intrusion Detection
- M16.14 Secure Wi-Fi Configuration & Hardening
- M16.15 Wireless Attack Detection & Countermeasures

MODULE 17 - MOBILE APPLICATION & MOBILE DEVICE SECURITY

18 lessons

- M17.1 Android & iOS Security Fundamentals
- M17.2 Mobile Application Architecture
- M17.3 OWASP Mobile Application Security Risks
- M17.4 Android Debug Bridge - ADB Security Testing
- M17.5 Static Android Analysis with JADX
- M17.6 APK Analysis with APKTool
- M17.7 Mobile Security Framework - MobSF
- M17.8 Runtime Mobile Application Testing with Frida

- M17.9 Android Permissions & Application Security
- M17.10 Android Rooting Security Concepts
- M17.11 iOS Jailbreaking Security Concepts
- M17.12 Mobile Phishing Threats
- M17.13 OTP & Mobile Authentication Threats
- M17.14 Mobile Malware & Malicious Applications
- M17.15 Google Play Protect & Mobile Application Protection
- M17.16 Mobile Device Management - MDM
- M17.17 Enterprise Mobile Security
- M17.18 Mobile Device Hardening & Countermeasures

MODULE 18 - IoT, IIoT, OT, ICS & SCADA SECURITY

19 lessons

- M18.1 Introduction to IoT Security
- M18.2 IoT Architecture & Technologies
- M18.3 IoT Protocols
- M18.4 IoT Vulnerabilities & Threats
- M18.5 OWASP IoT Security Risks
- M18.6 IoT Device Discovery
- M18.7 IoT Traffic Capture & Analysis
- M18.8 MQTT Security
- M18.9 Firmware Security Fundamentals
- M18.10 Firmware Analysis with Binwalk
- M18.11 Introduction to Industrial IoT - IIoT
- M18.12 Introduction to Operational Technology - OT
- M18.13 ICS & SCADA Security Fundamentals
- M18.14 Modbus & Industrial Protocol Security
- M18.15 OT Threats & Vulnerabilities
- M18.16 Critical Infrastructure Security
- M18.17 Network Segmentation for OT Environments
- M18.18 IoT/OT Attack Detection
- M18.19 IoT/OT Defensive Countermeasures

MODULE 19 - CLOUD, CONTAINER & KUBERNETES SECURITY

24 lessons

- M19.1 Cloud Computing Fundamentals
- M19.2 Cloud Service Models - IaaS, PaaS & SaaS
- M19.3 Public, Private & Hybrid Cloud
- M19.4 Shared Responsibility Model
- M19.5 Fog Computing & Security
- M19.6 Edge Computing & Security
- M19.7 AWS Security Fundamentals
- M19.8 Microsoft Azure Security Fundamentals
- M19.9 Google Cloud Security Fundamentals
- M19.10 Cloud IAM Security
- M19.11 Cloud Permissions & Privilege Management

- M19.12 Cloud Storage & S3 Misconfiguration Security
- M19.13 Cloud Network Security
- M19.14 Docker Fundamentals & Container Security
- M19.15 Docker Image Vulnerability Assessment
- M19.16 Container Vulnerability Assessment with Trivy
- M19.17 Kubernetes Security Fundamentals
- M19.18 Kubernetes Threats & Vulnerabilities
- M19.19 Kubernetes RBAC & Access Security
- M19.20 Serverless Computing Security
- M19.21 Cloud Logging & Monitoring
- M19.22 Cloud Security Posture & Misconfiguration Management
- M19.23 Cloud Security Controls & Countermeasures
- M19.24 AI-Assisted Cloud Security Assessment

MODULE 20 - CRYPTOGRAPHY & DATA SECURITY

21 lessons

- M20.1 Introduction to Cryptography
- M20.2 Classical Ciphers & Cryptography History
- M20.3 Symmetric Encryption
- M20.4 Asymmetric Encryption
- M20.5 Hashing & Message Digests
- M20.6 Password Hashing & Secure Password Storage
- M20.7 Digital Signatures
- M20.8 Public Key Infrastructure - PKI
- M20.9 Digital Certificates
- M20.10 OpenSSL Practical
- M20.11 TLS/SSL Security
- M20.12 GPG Email & File Encryption
- M20.13 Disk Encryption & Data-at-Rest Protection
- M20.14 Encryption in Transit vs Encryption at Rest
- M20.15 Cryptanalysis Concepts
- M20.16 Blockchain Fundamentals
- M20.17 Blockchain Security
- M20.18 Quantum Computing Security Risks
- M20.19 Quantum Cryptography Concepts
- M20.20 Post-Quantum Cryptography
- M20.21 AI-Assisted Cryptography & Cryptanalysis

PART II - PROFESSIONAL CYBERSECURITY TRACK

Extended coverage for cybersecurity governance, secure infrastructure, identity, endpoint defense, SOC/SIEM, threat intelligence and hunting, incident response and digital forensics, resilience, DevSecOps, AI security, purple teaming, capstones, reporting, and career development.

MODULE 21 - CYBERSECURITY GOVERNANCE, RISK & COMPLIANCE

17 lessons

- M21.1 Introduction to Cybersecurity Governance
- M21.2 NIST Cybersecurity Framework - CSF 2.0
- M21.3 Govern, Identify, Protect, Detect, Respond & Recover
- M21.4 Cybersecurity Policies, Standards, Procedures & Guidelines
- M21.5 Asset Management & Asset Inventory
- M21.6 Data Classification
- M21.7 Risk Identification & Risk Assessment
- M21.8 Risk Treatment - Mitigate, Transfer, Avoid & Accept
- M21.9 Risk Register Fundamentals
- M21.10 Security Control Selection
- M21.11 Security Auditing Fundamentals
- M21.12 Compliance & Regulatory Fundamentals
- M21.13 ISO/IEC 27001 Overview
- M21.14 NIST Security Standards & Guidelines Overview
- M21.15 PCI DSS, Privacy & Data Protection Concepts
- M21.16 Third-Party & Supply-Chain Risk
- M21.17 Cybersecurity Metrics, KPIs & Reporting

MODULE 22 - NETWORK SECURITY & SECURE INFRASTRUCTURE

22 lessons

- M22.1 Network Security Fundamentals
- M22.2 Secure Network Architecture
- M22.3 Network Segmentation & Microsegmentation
- M22.4 VLAN Security
- M22.5 Access Control Lists - ACL
- M22.6 Firewall Architecture & Rule Management
- M22.7 Stateful vs Stateless Firewalls
- M22.8 Next-Generation Firewalls - NGFW
- M22.9 pfSense Firewall Fundamentals
- M22.10 Fortinet/FortiGate Security Concepts
- M22.11 VPN Fundamentals
- M22.12 IPsec VPN Security
- M22.13 SSL/TLS VPN Security
- M22.14 DNS Security
- M22.15 DHCP Security
- M22.16 Network Access Control
- M22.17 Secure Remote Access
- M22.18 Proxy, Reverse Proxy & Secure Web Gateway

- M22.19 Network Monitoring
- M22.20 Network Baselines & Anomaly Detection
- M22.21 Zero Trust Network Architecture
- M22.22 Secure Network Configuration & Hardening

MODULE 23 - IDENTITY & ACCESS MANAGEMENT

18 lessons

- M23.1 Identity & Access Management - IAM Fundamentals
- M23.2 Identification, Authentication & Authorization
- M23.3 Role-Based Access Control - RBAC
- M23.4 Attribute-Based Access Control - ABAC
- M23.5 Principle of Least Privilege
- M23.6 Separation of Duties
- M23.7 Multi-Factor Authentication - MFA
- M23.8 Password Policies & Passwordless Authentication
- M23.9 Single Sign-On - SSO
- M23.10 Active Directory Identity Security
- M23.11 Kerberos Security Fundamentals
- M23.12 Group Policy Security
- M23.13 Privileged Access Management - PAM
- M23.14 Service Account Security
- M23.15 Account Provisioning & Deprovisioning
- M23.16 Cloud IAM Security
- M23.17 Identity Threat Detection & Response
- M23.18 Active Directory Hardening

MODULE 24 - WINDOWS, LINUX & ENDPOINT SECURITY

22 lessons

- M24.1 Endpoint Security Fundamentals
- M24.2 Windows Security Architecture
- M24.3 Windows Security Hardening
- M24.4 Windows Defender & Microsoft Defender Fundamentals
- M24.5 Windows Firewall Configuration & Security
- M24.6 Windows Event Logs
- M24.7 Sysmon for Security Monitoring
- M24.8 Windows Update & Patch Management
- M24.9 BitLocker & Windows Data Protection
- M24.10 Linux Security Architecture
- M24.11 Linux User, Group & Permission Security
- M24.12 Linux File Permission Hardening
- M24.13 SSH Security & Hardening
- M24.14 Linux Firewall Security
- M24.15 Linux Logging & Monitoring
- M24.16 Linux Patch & Package Security
- M24.17 Antivirus, EDR & XDR
- M24.18 Host Intrusion Detection

- M24.19 Application Allowlisting & Execution Control
- M24.20 Endpoint Security Baselines
- M24.21 CIS Security Benchmark Concepts
- M24.22 Endpoint Detection, Investigation & Response

MODULE 25 - SOC, SIEM & SECURITY MONITORING

24 lessons

- M25.1 Security Operations Center - SOC Fundamentals
- M25.2 SOC Analyst Roles - Tier 1, Tier 2 & Tier 3
- M25.3 Security Event vs Alert vs Incident
- M25.4 Logging Fundamentals
- M25.5 Windows Security Logs
- M25.6 Linux Security Logs
- M25.7 Firewall, IDS & Application Logs
- M25.8 SIEM Fundamentals
- M25.9 SIEM Architecture & Log Collection
- M25.10 Wazuh Security Monitoring
- M25.11 Splunk Fundamentals for Cybersecurity
- M25.12 Elastic Security Fundamentals
- M25.13 Security Onion Fundamentals
- M25.14 Alert Triage & Investigation
- M25.15 Correlation Rules & Detection Logic
- M25.16 MITRE ATT&CK Mapping for Defenders
- M25.17 Sigma Detection Rules Fundamentals
- M25.18 IOC-Based Detection
- M25.19 Behavior-Based Detection
- M25.20 Network Security Monitoring
- M25.21 Endpoint Security Monitoring
- M25.22 Building a SOC Investigation Workflow
- M25.23 AI-Assisted SOC Alert Analysis
- M25.24 AI for Security Operations & Alert Prioritization

MODULE 26 - CYBER THREAT INTELLIGENCE & THREAT HUNTING

20 lessons

- M26.1 Cyber Threat Intelligence Fundamentals
- M26.2 Strategic, Operational, Tactical & Technical Intelligence
- M26.3 Threat Actors & Adversary Profiles
- M26.4 Tactics, Techniques & Procedures - TTPs
- M26.5 Indicators of Compromise - IOC
- M26.6 Indicators of Attack - IOA
- M26.7 MITRE ATT&CK for Threat Intelligence
- M26.8 Threat Intelligence Sources & Feeds
- M26.9 MISP Threat Intelligence Platform Fundamentals
- M26.10 STIX & TAXII Fundamentals
- M26.11 YARA Rules Fundamentals
- M26.12 Sigma Rules Fundamentals

- M26.13 Introduction to Threat Hunting
- M26.14 Hypothesis-Driven Threat Hunting
- M26.15 Endpoint Threat Hunting
- M26.16 Network Threat Hunting
- M26.17 Hunting with Logs & SIEM
- M26.18 Threat Hunting with MITRE ATT&CK
- M26.19 AI-Assisted Threat Intelligence
- M26.20 AI-Assisted Threat Hunting

MODULE 27 - INCIDENT RESPONSE & DIGITAL FORENSICS

24 lessons

- M27.1 Cybersecurity Incident Response Fundamentals
- M27.2 Incident Response Roles & Responsibilities
- M27.3 Incident Detection & Analysis
- M27.4 Incident Classification & Severity
- M27.5 Incident Containment
- M27.6 Eradication & Recovery
- M27.7 Post-Incident Lessons Learned
- M27.8 Incident Response Playbooks
- M27.9 Phishing Incident Response
- M27.10 Malware Incident Response
- M27.11 Ransomware Incident Response
- M27.12 Account Compromise Response
- M27.13 Data Breach Response Fundamentals
- M27.14 Introduction to Digital Forensics
- M27.15 Evidence Handling & Chain of Custody
- M27.16 Disk Forensics Fundamentals
- M27.17 Memory Forensics Fundamentals
- M27.18 Network Forensics Fundamentals
- M27.19 Autopsy Digital Forensics Fundamentals
- M27.20 Volatility Memory Analysis Fundamentals
- M27.21 Timeline Analysis
- M27.22 Log Forensics
- M27.23 Writing an Incident Report
- M27.24 AI-Assisted Incident Analysis

MODULE 28 - DATA SECURITY, BACKUP, DISASTER RECOVERY & BUSINESS CONTINUITY

20 lessons

- M28.1 Data Security Fundamentals
- M28.2 Data Classification & Handling
- M28.3 Data-at-Rest Security
- M28.4 Data-in-Transit Security
- M28.5 Data-in-Use Security
- M28.6 Data Loss Prevention - DLP
- M28.7 Secure File Sharing & Storage

- M28.8 Backup Fundamentals
- M28.9 Full, Incremental & Differential Backups
- M28.10 3-2-1 Backup Strategy
- M28.11 Offline & Immutable Backups
- M28.12 Backup Security Against Ransomware
- M28.13 Backup Restoration & Testing
- M28.14 Recovery Point Objective - RPO
- M28.15 Recovery Time Objective - RTO
- M28.16 Disaster Recovery Fundamentals
- M28.17 Disaster Recovery Planning
- M28.18 Business Continuity Planning
- M28.19 High Availability, Redundancy & Failover
- M28.20 Cyber Resilience

MODULE 29 - SECURE DEVELOPMENT, DEVSECOPS & SUPPLY-CHAIN SECURITY

20 lessons

- M29.1 Secure Software Development Lifecycle - SSDLC
- M29.2 Security Requirements & Threat Modeling
- M29.3 Secure Coding Fundamentals
- M29.4 Input Validation & Output Encoding
- M29.5 Authentication & Authorization Design
- M29.6 Secrets & API Key Management
- M29.7 Static Application Security Testing - SAST
- M29.8 Dynamic Application Security Testing - DAST
- M29.9 Software Composition Analysis - SCA
- M29.10 Dependency & Package Security
- M29.11 DevSecOps Fundamentals
- M29.12 CI/CD Pipeline Security
- M29.13 Git & Source Code Repository Security
- M29.14 Container Pipeline Security
- M29.15 Infrastructure as Code Security
- M29.16 Software Bill of Materials - SBOM
- M29.17 Software Supply-Chain Security
- M29.18 Third-Party Dependency Risk
- M29.19 Secrets Scanning
- M29.20 Security Testing Automation

MODULE 30 - AI & MACHINE LEARNING SECURITY

20 lessons

- M30.1 Artificial Intelligence Fundamentals for Cybersecurity
- M30.2 Machine Learning Fundamentals for Security Professionals
- M30.3 Generative AI & Large Language Models
- M30.4 AI in Offensive Security
- M30.5 AI in Defensive Security
- M30.6 AI-Assisted Reconnaissance

- M30.7 AI-Assisted Vulnerability Analysis
- M30.8 AI-Assisted Security Monitoring
- M30.9 AI-Assisted Incident Response
- M30.10 AI-Assisted Threat Intelligence
- M30.11 AI Security Risks
- M30.12 Prompt Injection Security Concepts
- M30.13 Sensitive Data Leakage Through AI Systems
- M30.14 AI Model & Training Data Security
- M30.15 Adversarial Machine Learning Concepts
- M30.16 Deepfake Security Threats
- M30.17 AI Governance & Responsible AI
- M30.18 Secure Enterprise Use of Generative AI
- M30.19 AI Security Monitoring & Controls
- M30.20 Future of AI in Cybersecurity

MODULE 31 - PENETRATION TESTING, PURPLE TEAM & SECURITY ASSESSMENT

15 lessons

- M31.1 Building an Ethical Hacking Methodology
- M31.2 Penetration Testing Methodology
- M31.3 Rules of Engagement
- M31.4 Scoping a Security Assessment
- M31.5 Black-Box, Gray-Box & White-Box Testing
- M31.6 Vulnerability Assessment vs Penetration Testing
- M31.7 Red Team Fundamentals
- M31.8 Blue Team Fundamentals
- M31.9 Purple Team Fundamentals
- M31.10 Mapping Security Tests to MITRE ATT&CK
- M31.11 Detection Validation
- M31.12 Attack Simulation & Defensive Validation
- M31.13 Evidence Collection
- M31.14 Writing Technical Findings
- M31.15 Writing Remediation Recommendations

MODULE 32 - CAPSTONE, REPORTING, EXAM PREPARATION & CAREER DEVELOPMENT

32 lessons

- M32.1 Complete Ethical Hacking Engagement Workflow
- M32.2 Complete Cybersecurity Assessment Workflow
- M32.3 Vulnerability Assessment Capstone
- M32.4 Web Application Security Assessment Capstone
- M32.5 Network Security Assessment Capstone
- M32.6 SOC Investigation Capstone
- M32.7 Incident Response Capstone
- M32.8 Purple Team Attack & Detection Validation Capstone
- M32.9 Writing a Professional Vulnerability Assessment Report

- M32.10 Writing a Professional Penetration Testing Report
- M32.11 Writing a Cybersecurity Assessment Report
- M32.12 Writing an Incident Response Report
- M32.13 Executive Summary Writing for Management
- M32.14 Vulnerability Evidence & Proof-of-Concept Documentation
- M32.15 CVSS Scoring & Vulnerability Prioritization Practical
- M32.16 Risk-Based Remediation Planning
- M32.17 Building Your Own Cybersecurity Home Lab
- M32.18 Capture-the-Flag Platforms & Practical Skill Development
- M32.19 Building a Cybersecurity Portfolio
- M32.20 Building GitHub Cybersecurity Projects
- M32.21 How to Continue Cybersecurity Research
- M32.22 Reading CVEs, Security Advisories & Technical Research
- M32.23 CEH v13 / CEH AI Knowledge Exam Preparation
- M32.24 CEH Practical Exam Preparation Overview
- M32.25 Cybersecurity Interview Preparation
- M32.26 Cybersecurity Career Roadmap After CEH
- M32.27 Ethical Hacker / Penetration Tester Career Roadmap
- M32.28 SOC Analyst / Blue Team Career Roadmap
- M32.29 Network & Infrastructure Security Career Roadmap
- M32.30 Cloud Security Career Roadmap
- M32.31 Incident Response & DFIR Career Roadmap
- M32.32 Continuous Learning & Cybersecurity Certification Roadmap

Alignment References & Authoritative Sources

These references support the curriculum alignment and terminology. Product/tool-specific lessons should also be updated against current vendor documentation when the course is recorded or revised.

EC-Council - Certified Ethical Hacker (CEH AI)

<https://www.eccouncil.org/train-certify/certified-ethical-hacker-ceh/>

Official CEH program page describing the current AI-enhanced CEH structure, 20 learning modules, labs, attack techniques and learning framework.

EC-Council - CEH v13

<https://ethicalhacking.eccouncil.org/>

Official CEH v13 information highlighting AI integration, hands-on labs, the Learn/Certify/Engage/Compete framework and current program positioning.

NIST - Cybersecurity Framework (CSF) 2.0

<https://www.nist.gov/publications/nist-cybersecurity-framework-csf-20>

Authoritative cybersecurity risk-management framework organized around Govern, Identify, Protect, Detect, Respond and Recover.

MITRE ATT&CK - Enterprise Tactics

<https://attack.mitre.org/tactics/>

Authoritative knowledge base for adversary tactics and techniques used throughout threat intelligence, detection engineering, threat hunting and purple teaming.

OWASP - API Security Project

<https://owasp.org/projects/api-security-project>

Authoritative community resource for API security risks and secure API practices, including the OWASP API Security Top 10.

Document Notes

Total structured lesson topics: 587. The document uses a course-outline format so each item can be produced as a dedicated video, lecture, lab, workshop, or combined session depending on delivery depth.

Recommended maintenance cycle: review the syllabus at least every 6-12 months for changes in threat techniques, security standards, cloud platforms, AI-security guidance, OWASP projects, MITRE ATT&CK, and vendor tooling.